Nonbreeding site use of continental black-tailed godwits in the East Atlantic Flyway (2013-2025)

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Executive summary

This report presents an assessment of nonbreeding site use by continental black-tailed godwits (*Limosa limosa limosa*) across the East Atlantic Flyway, based on satellite tracking data collected between 2013 and 2025. The analysis includes 347 individuals (946 individual-years and >240,000 locations). The objective is to identify core nonbreeding sites, quantify their relative importance, define peak-use windows, and provide a baseline for conservation planning and evaluation under the projects LIFE GrassBirdHabitats and Godwit Flyway.

Key findings:

1. **Nonbreeding use is concentrated across 12 sites**
   - A small number of sites support a large share of tracked birds.
   - Doñana supports on average 64% of individuals during the nonbreeding season.
   - The Senegal River Delta supports 48%, particularly immediately after southward migration.
   - Several additional sites consistently support 15-40% of tracked birds.

2. Seasonal redistribution follows a clear pattern
Three distinct phases occur annually:
- July-August: Peak use of coastal West African sites after arrival from breeding areas.
- October-January: Redistribution to Sahelian floodplain systems, including the Lower Senegal River and Inner Niger Delta.
- February-March: Peak use of Iberian sites during late-winter staging prior to spring migration.

3. Interannual patterns are stable
Although annual variation occurs, the relative importance of major sites and the overall seasonal structure remain broadly consistent from 2013 to 2025. Extremadura is an exception, showing a strong decline in use after the first two years of data availability.

Background and aims

Godwits breeding in northwestern Europe migrate along the East Atlantic Flyway to sites in Iberia and West Africa. During this period, birds concentrate at a limited number of wetlands, estuaries, and floodplain systems. Several of these nonbreeding areas are subject to ongoing wetland loss and degradation (Howison et al. 2019; Vansteelant 2023; Craft et al., 2025). At the same time, restoration and management measures are being implemented at key sites under LIFE projects and related conservation initiatives. Effective prioritisation and evaluation of these efforts requires a clear understanding of where birds concentrate, when peak use occurs, and how consistent site use patterns are over time.

Previous assessments of nonbreeding distribution were based primarily on earlier tracking technologies and smaller datasets (Hooijmeijer et al. 2014; Verhoeven et al. 2021). While these

studies identified important regions, we still lack a consistent, multi-year quantification of relative site importance and seasonal timing across the flyway.

The main objectives of this report are to:

- Provide an overview of key nonbreeding sites across the EAF
- Quantify their relative importance based on the proportion of tracked birds using each site
- Identify peak periods of site use during the nonbreeding season
- Establish a baseline against which future changes in site use can be evaluated

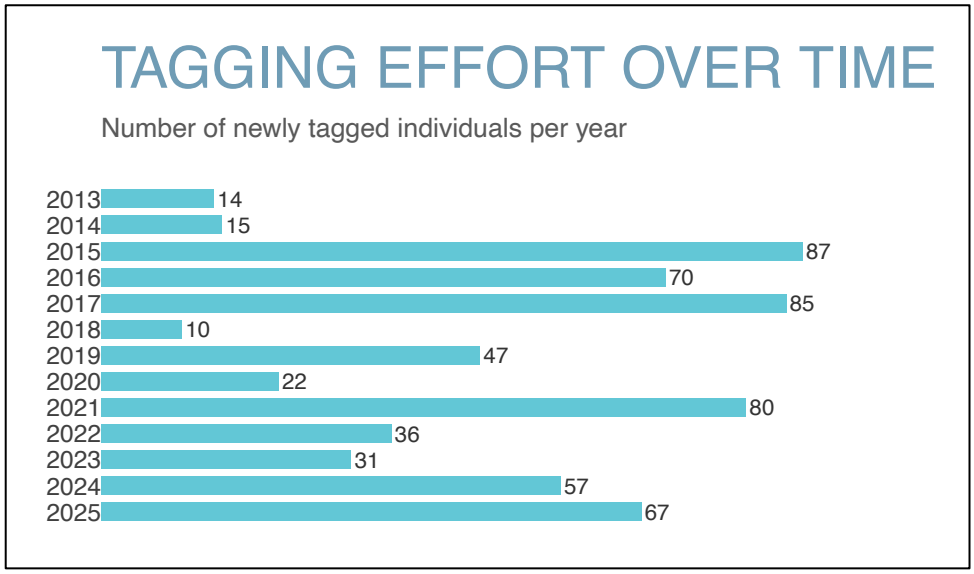
This baseline is particularly relevant for monitoring the effects of habitat restoration measures implemented at several sites within the framework of the LIFE projects GrassBirdHabitats and Godwit Flyway, which aim to improve godwit habitat quality and long-term conservation outcomes across the flyway. In addition, understanding peak timing patterns can help prioritise field survey efforts and optimize coordinated counts across sites among research partners.

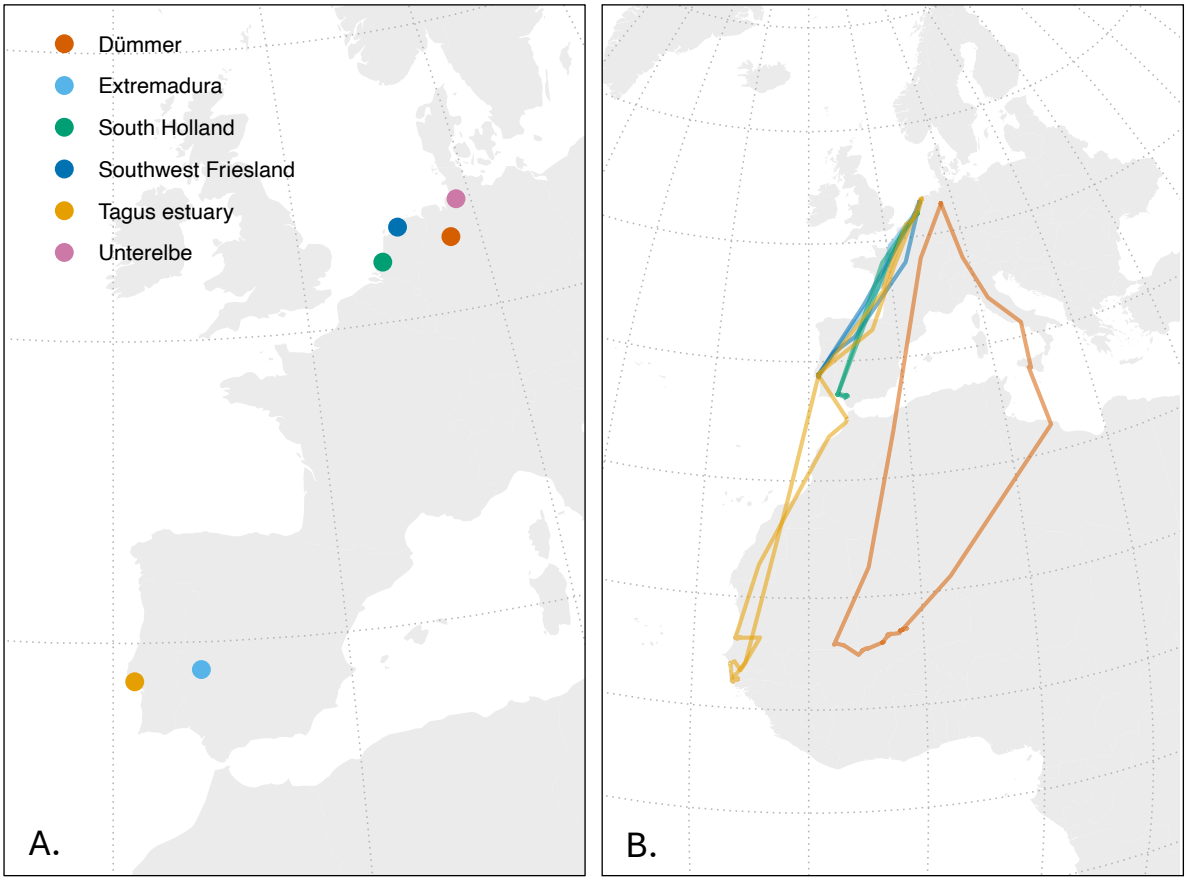
Data overview and analysis

1. Tracking dataset
Birds included in this analysis were tagged with satellite tags (PTT and GPS) in sites that represent major breeding and staging areas within the northwestern European population and provide strong coverage of the main European migratory corridor. In total, 540 individuals were initially available. After filtering and standardisation (see below), 347 individuals were retained for analysis, representing 946 individual-year combinations and 240,392 locations. These data provide the most extensive assessment of nonbreeding site use for this population to date.

Overview of tagging sites and numbers of tagged godwits.

Site	Country	Individuals (raw)	Individuals (filtered)
Southwest Friesland	NL	308	166
South Holland	NL	9	8
Dümmer	DE	80	56
Unterelbe	DE	2	1
Extremadura	ES	75	68
Tagus estuary	PT	66	48





Tagging locations (A) and example migration routes (B) of tracked godwits along the EAF.

2. Data filtering and standardisation

To ensure comparability across individuals and years, the following steps were applied:

- Removed all in-flight locations (ground speed > 0)
- Removed non-migratory individuals (total latitudinal change < 5°)
- Restricted data to latitude < 42° for rough exclusion of breeding sites
- Subsampled to four locations per individual per day to reduce bias of high frequency tags
- Defined the nonbreeding season as beginning on 1 June, corresponding to the annual minimum occupancy point in the nonbreeding range

3. Identifying core nonbreeding sites

Core nonbreeding sites were identified using DBSCAN (Density-Based Spatial Clustering of Applications with Noise), which groups spatial points based on local density. DBSCAN identifies areas where locations are densely aggregated and treats isolated points as noise. This approach allows sites to emerge from the data without arbitrary predefined boundaries and is well suited for identifying regional sites used repeatedly by many individuals. In this analysis, a clustering radius of 30 km was used with a minimum of 1,500 fixes required for a cluster to qualify as a site.

4. Site-level metrics

For each site, two main temporal metrics were calculated and averaged across years:

1. Peak window: the 14-day period with the highest number of tagged birds present
2. Proportion of tracked birds: the proportion of tagged individuals using the site during the nonbreeding season

Core nonbreeding sites identified

Using the clustering approach described above, 12 core nonbreeding sites were identified, representing areas of repeated and concentrated use by multiple individuals across years.

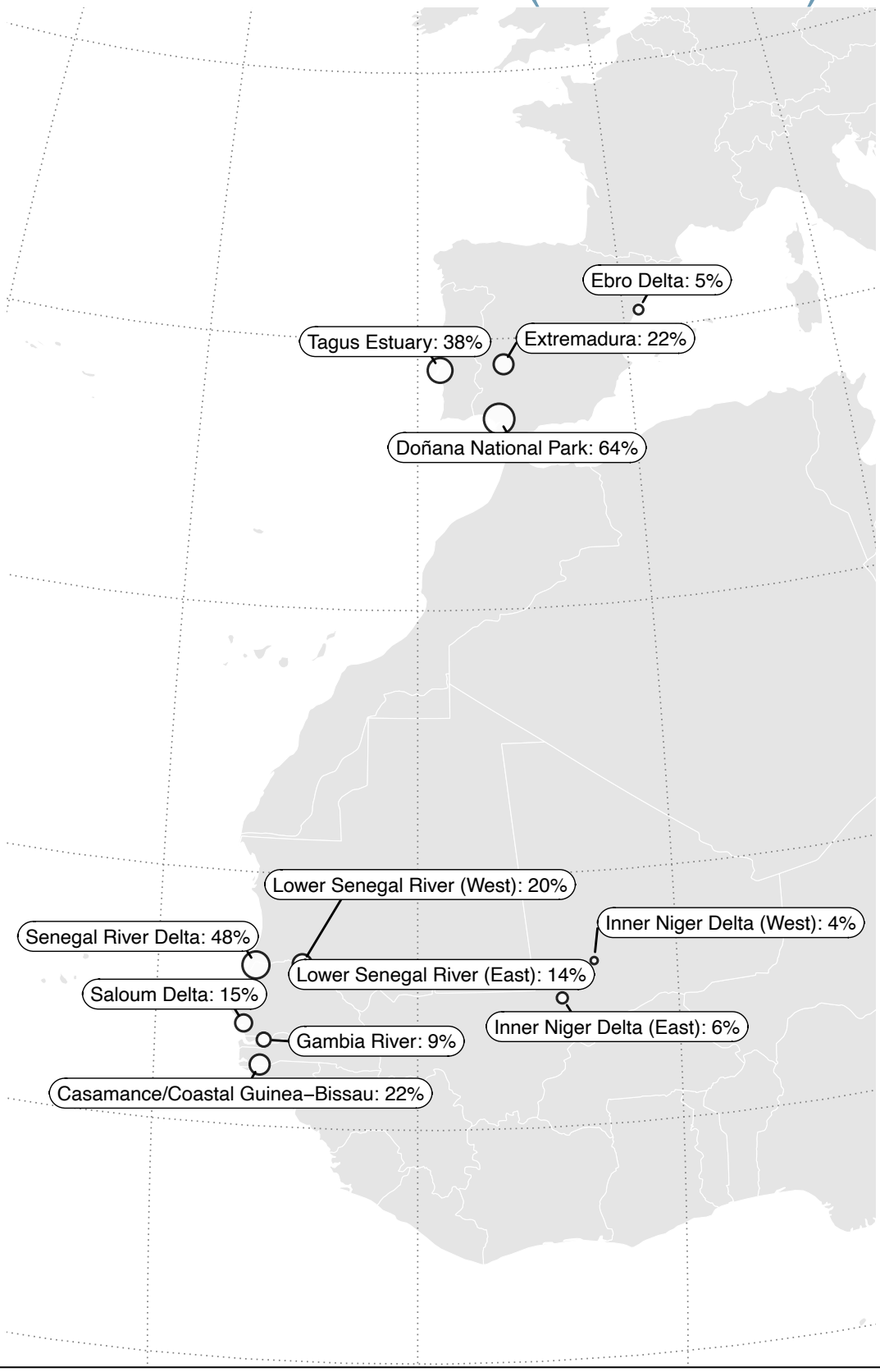
1. Spatial distribution of core sites
Core sites are distributed across three main regions:

- Iberia: Doñana, Tagus Estuary, Extremadura, Ebro Delta
- Coastal West Africa: Senegal River Delta, Saloum Delta, Casamance/Coastal Guinea-Bissau, Gambia River
- Sahelian floodplain systems: Lower Senegal River (East/West), Inner Niger Delta (East/West)

Non-breeding site use of godwits across the East Atlantic Flyway.

Site	Proportion of tracked birds	Peak window
Doñana National Park	0.64	Jul 08-Jul 21
Senegal River Delta	0.48	Jul 18-Jul 31
Tagus Estuary	0.38	Feb 17-Mar 02
Casamance / Coastal Guinea-Bissau	0.22	Aug 02-Aug 15
Extremadura	0.22	Feb 14-Feb 27
Lower Senegal River (West)	0.20	Oct 17-Oct 30
Saloum Delta	0.15	Jul 30-Aug 12
Lower Senegal River (East)	0.14	Oct 30-Nov 12
Gambia River	0.09	Oct 15-Oct 28
Inner Niger Delta (East)	0.06	Jan 03-Jan 16
Ebro Delta	0.05	Mar 09-Mar 22
Inner Niger Delta (West)	0.04	Nov 10-Nov 23

CORE NONBREEDING SITES AND AVERAGE SITE USE (2013–2025)

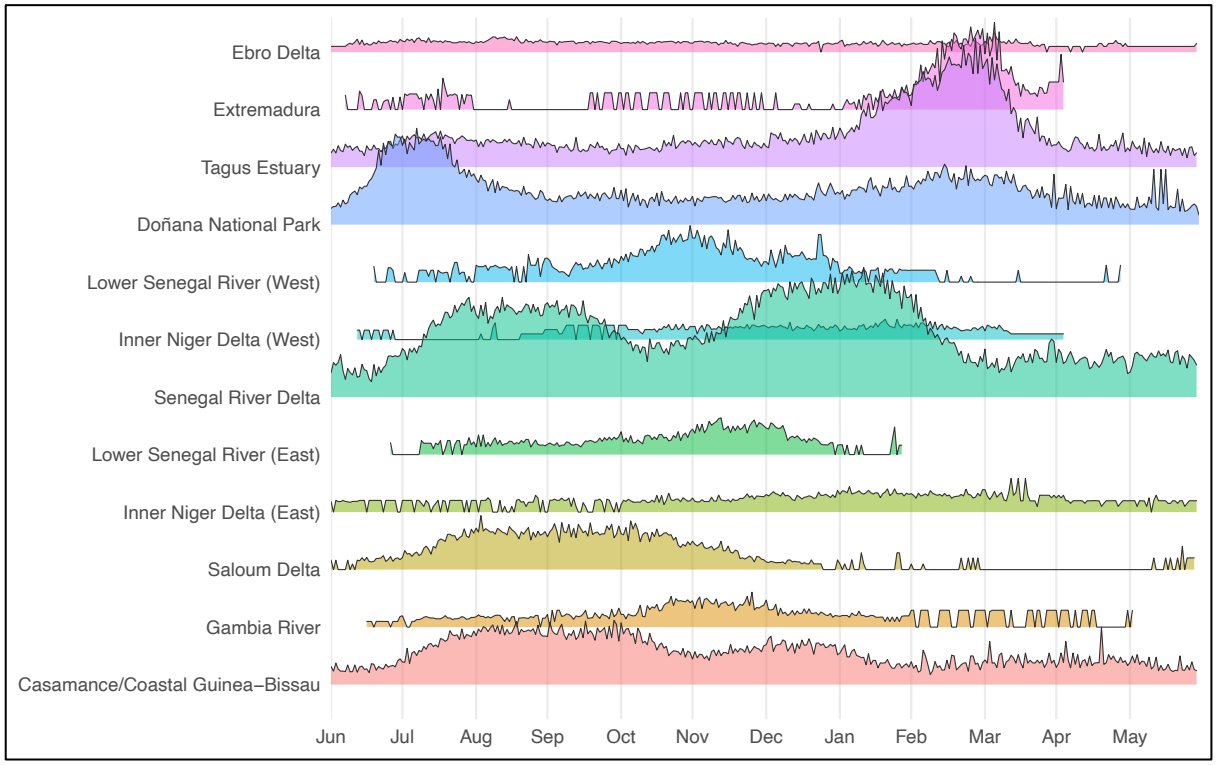


Core non-breeding sites of godwits across the East Atlantic Flyway. Percentages indicate the proportion of tagged individuals using each site during the non-breeding season, averaged across 2013 to 2025.

2. Seasonal redistribution patterns

Daily occupancy patterns reveal a structured seasonal redistribution across regions. Three phases are consistently observed:

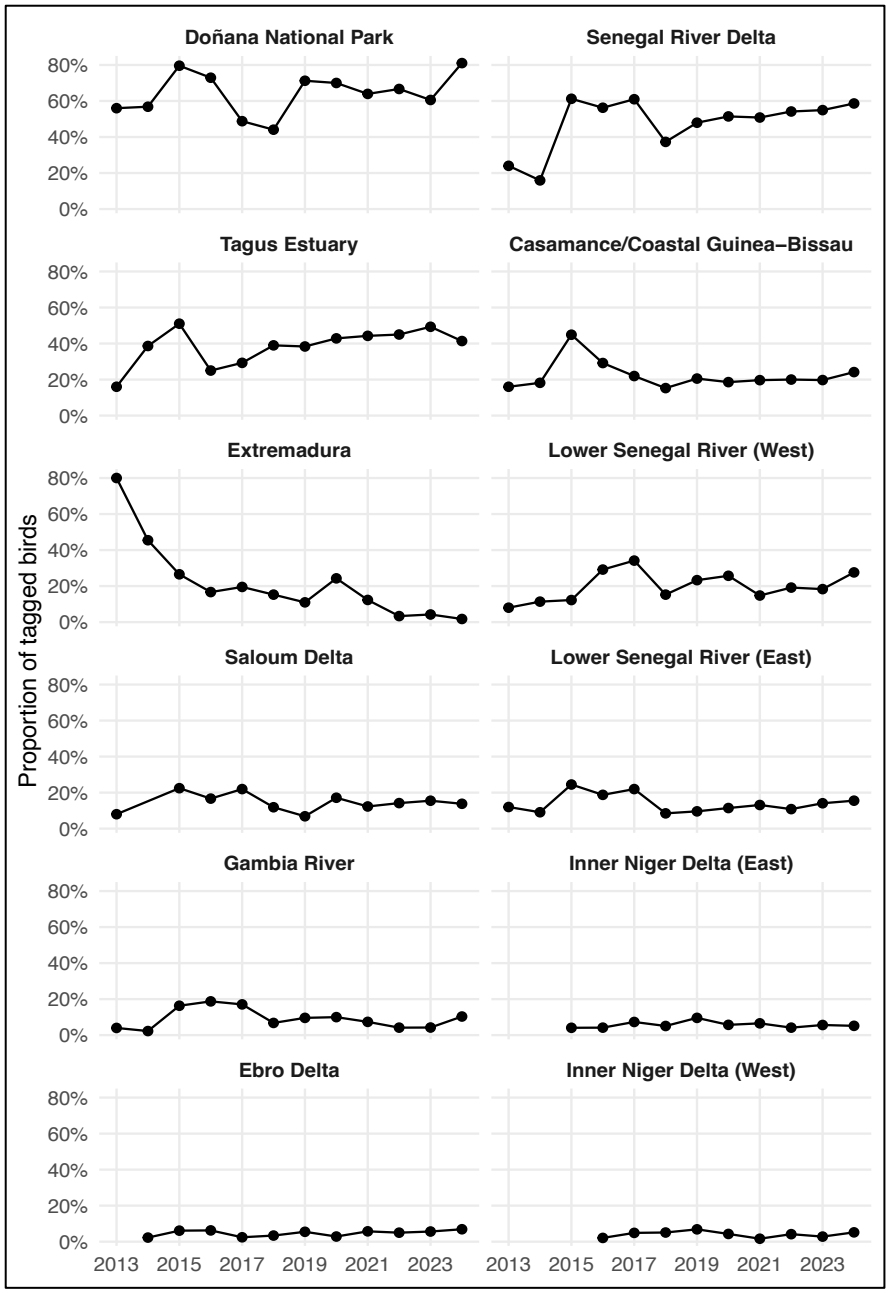
- **Early nonbreeding (July-August):** Peak use of coastal West African sites, particularly the Senegal River Delta and Saloum Delta.
- **Mid nonbreeding (October-January):** Increased use of inland Sahel floodplain systems, including Lower Senegal River and Inner Niger Delta.
- **Late nonbreeding (February-March):** Peak use of Iberian sites, particularly the Tagus Estuary and Extremadura, prior to northward migration.



Seasonal ridgeline plot showing daily proportion of tracked individuals per site, averaged across years. Sites ordered by latitude (north to south).

3. Interannual Consistency

Although annual variation occurs, the relative importance of most major sites and the overall seasonal structure remain broadly consistent from 2013 to 2025. Extremadura is a notable exception, showing a marked decline in use over time.



Annual variation in non-breeding site use from 2013 to 2025.

Conclusions

Nonbreeding site use of godwits along the East Atlantic Flyway is concentrated and seasonally structured. A limited number of wetland sites support a large proportion of tracked individuals, with Doñana and the Senegal River Delta consistently emerging as the most important sites.

Use of sites follows a predictable annual pattern, shifting from coastal West Africa in early nonbreeding, to Sahelian floodplains in mid-season, and to Iberian staging areas prior to spring migration. The results provide a quantitative baseline for prioritising conservation action, coordinating flyway-scale monitoring, and evaluating the outcomes of habitat restoration measures.

For custom godwit site-use metrics, explore tracks interactively with the Flyway Movement Viewer tool at: https://tbcraft.shinyapps.io/flyway_tracking_summary_app/

References

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